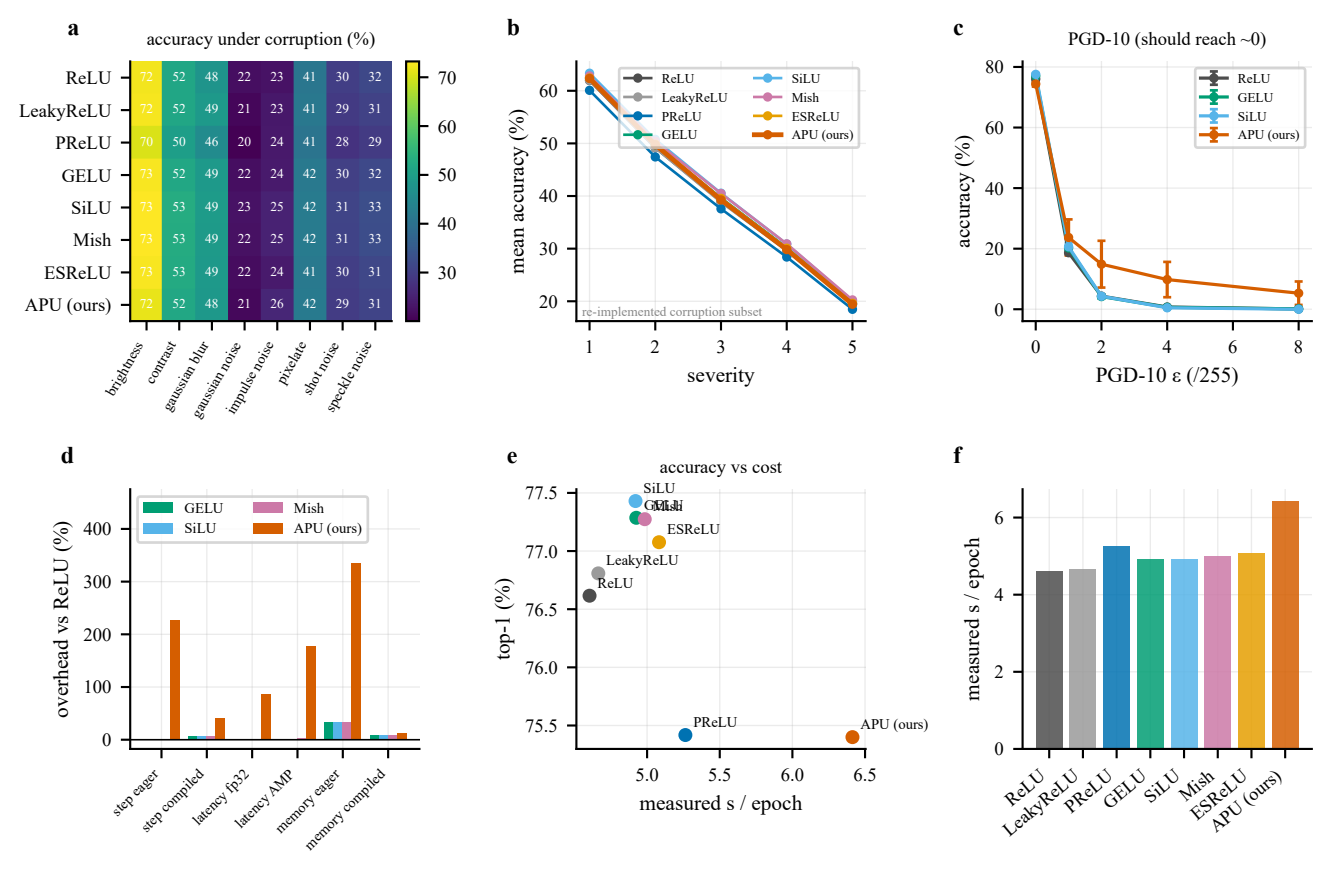

a
accuracy under corruption (%)
ReLU 72 52 48 22 23 41 30 32
LeakyReLU 72 52 49 21 23 41 29 31
PReLU 70 50 46 20 24 41 28 29
GELU 73 52 49 22 24 42 30 32
SiLU 73 53 49 23 25 42 31 33
Mish 73 53 49 22 25 42 31 33
ESReLU 73 53 49 22 24 41 30 31
APU (ours) 72 52 48 21 26 42 29 31
70
60
50
40
30
brightness
contrast
gaussian blur
gaussian noise
impulse noise
pixelate
shot noise
speckle noise
b
mean accuracy (%)
severity
re-implemented corruption subset
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)
c
PGD-10 (should reach ~0)
accuracy (%)
PGD-10 ε (/255)
ReLU
GELU
SiLU
APU (ours)
d
overhead vs ReLU (%)
GELU
SiLU
Mish
APU (ours)
step eager
step compiled
latency fp32
latency AMP
memory eager
memory compiled
e
accuracy vs cost
top-1 (%)
measured s / epoch
SiLU
ESReLU
LeakyReLU
ReLU
PReLU
APU (ours)
f
measured s / epoch
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)